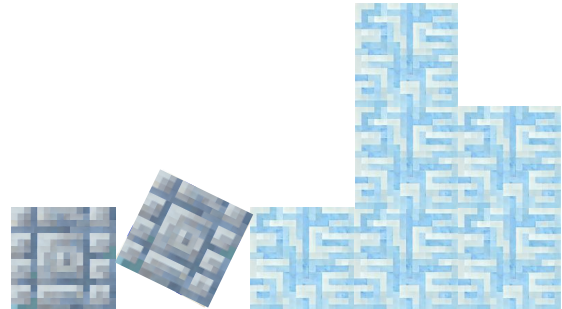


Mnetexture

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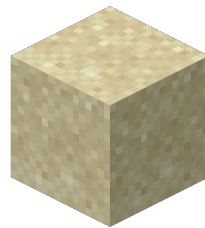
Project reminder

- From the game Minecraft
- Project goal: Generate items and blocks texture from a textual prompt

In-game screenshot



Sand block



Sand texture (1 side)



Sand block texture in different resource packs



[Halloween mash-up](#)



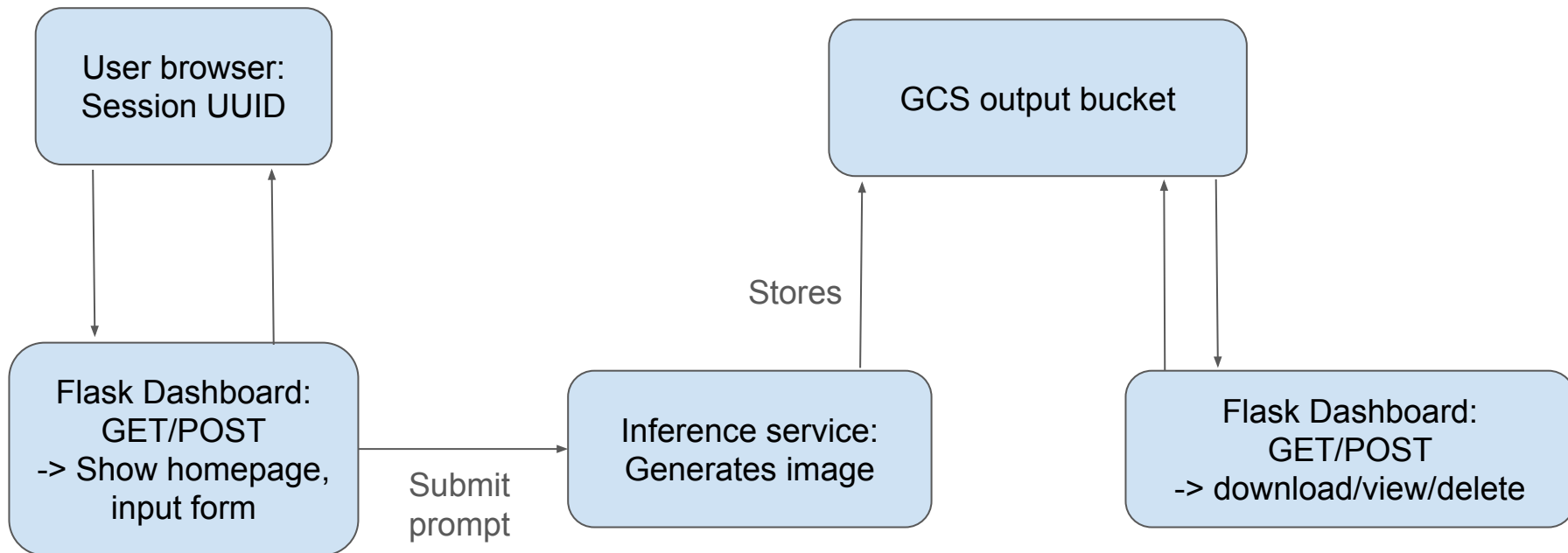
[Super-cute](#)



[Jolicraft](#)

Architecture overview

Utilisation flow



Flask Dashboard: Actions

Method	Route	Description
GET / POST	/	Homepage: Session init., view session / Submit prompt
GET	/image	Stream and display image directly from GCS bucket
GET	/download	Download generated texture as .png from GCS
POST	/delete	Delete image from GCS, update gallery

Deployment: 2 containers on Google cloud

Flask Dashboard

Base image: python:3.11-slim

Server: Gunicorn (8 threads)

Scale to 0

Inference Service

Base image:
nvidia/cuda:12.1.0-runtime

Server: Uvicorn (1 worker)

Compute: GPU required (nvidia-l4)

Scale to 0

Note: due to scale to 0, we have a cold start -> fetching files and loading the pipeline

Deployment: 2 containers on Google cloud

Flask Dashboard

Base image: python:3.11-slim

Server: Gunicorn (8 threads)

Scale to 0

Allow-unauthenticated

Inference Service

Base image:
nvidia/cuda:12.1.0-runtime

Server: Uvicorn (1 worker)

Compute: GPU required (nvidia-l4)

Scale to 0

Allow-unauthenticated

POST /generate includes the header X-API-Key : <key>
-> HTTP 403 Forbidden

GCS Bucket: Image storage

gs://generated_data_minetexture/generated

session_A (UUID)/

minecraft-stone-20260401.png

pixel-grass-20260331.png

session_B (UUID)/

flower-blue-realistic-20250401.png

...

Problem:

- Clean/expired cache
- Lost images inside the bucket

Future considerations:

- User authentication
- Guest images -> deleted

Model pipeline - Inference optimisation

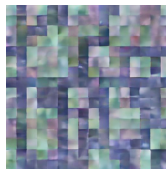
- Limit denoising step

- Less steps

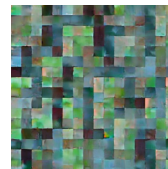
30 steps:
9min



20 steps:
5min50

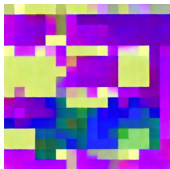


15 steps:
4min30

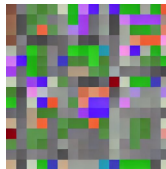


- Add LCM pretrained LoRA

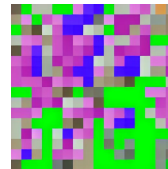
8 steps:



15 steps:



30 steps:



- Precision reducing (GPU)

32bits:
9sec



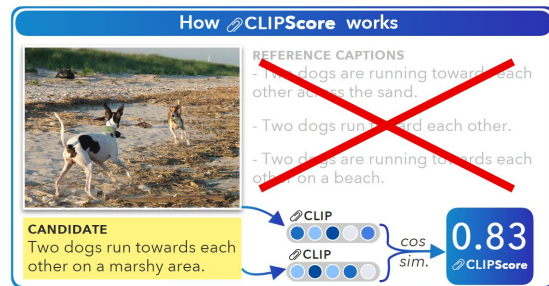
16bits:
3sec



Training and Deployment (update)



- Model .safetensor pushed to Google Cloud with name convention:
 - training_type/branch_name/commit_sha/timestamp
 - Single source of truth for training parameters: training_config.toml (at commit *commit_sha*)
- Updated training pipeline
 - Training/Test set: 93% of dataset for training set, 7% for test set.
 - Evaluation metric: average **CLIP score**.
- Training on-premise
 - Pushes the image to the *minetexture-checkpoints*
 - Used in our inference endpoint

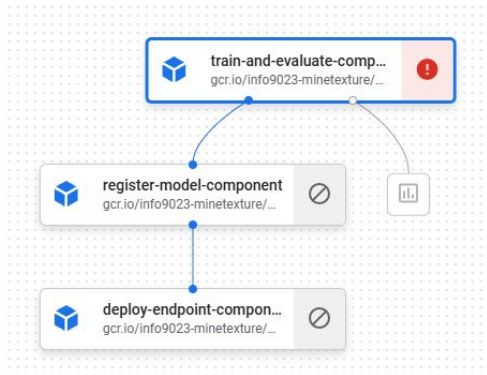


Training and Deployment in the cloud



Vertex AI

- GitHub Action Workflow runs when a push is made on a branch with name containing “*training*”
 - Builds the docker image for the training pipeline
 - Generates the KubeFlow pipeline
 - Starts the VertexAI training and outputs the training
- Pipeline:



- Google Cloud Secret Manager is used to stored *HF_TOKEN* and *WANDB_API_KEY*
- Remaining:
 - Finish VertexAI pipeline (*feature/training-v2* branch)

Demo

<https://dashboard-217219593769.europe-west1.run.app/>



Questions?

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